ZAIN ABBAS

Faisalabad, Pakistan — zainabbasm416@gmail.com — +92 346-4040192
linkedin.com/in/zainabbas102 — github.com/Zain-Abbas — zainabbas.me

SUMMARY

Electrical Engineering student at FAST NUCES (Class of 2027), CGPA 3.3, with hands-on experience in embedded systems, circuit design, automation, and software development. Teaching Assistant across three courses supporting 80+ students each. Skilled in hardware and software development with experience in Arduino, ESP32, Multisim, Proteus, MATLAB, C++, Python, and web technologies.

EDUCATION

FAST NUCES, Chiniot-Faisalabad Campus — Aug 2023 – Aug 2027
B.S. Electrical & Electronics Engineering — **CGPA: 3.3 / 4.0**

KIPS Education System — 2021 – 2023
FSC Pre-Engineering — **66%**

Dhariwal Public High School — 2019 – 2021
Matriculation, Science — **96%**

EXPERIENCE

Sadaqat Textile Limited — Jul 11, 2026 – Aug 13, 2026
Electrical Engineering Intern — Faisalabad, PK

- Gained hands-on industrial experience in electrical systems, machinery, maintenance, troubleshooting, and automation across textile manufacturing operations.

FAST NUCES — Feb 2025 – Present
Teaching Assistant — LCA, CVT & Differential Equations — Faisalabad, PK

- Supported 80+ students across 3 courses; conducted Multisim and Proteus labs, graded assignments, and provided technical mentorship.

CyberGen — Jul 2025 – Aug 2025
Intern — Artificial Intelligence — Faisalabad, PK

- Assisted with AI model development, data preprocessing, analysis, and applying machine learning concepts to practical problems.

ACADEMIC PROJECTS

- **IoT-Based Smart Home Gardening System** (ESP32, Sensors) — Real-time soil, temperature, and humidity monitoring with automated irrigation and Wi-Fi alerts. **[1st Place]**
- **User-Space Operating System Simulator** (C++, Ubuntu, VMware) — Simulated process scheduling, resource allocation, process handling, and system-level operations with real-time execution visualization.
- **Smart Home Automation System** (ATmega32) — Role-based access control with Admin (full control) and Guest (limited light control) modes.
- **Line Following Robot** (Arduino, IR Sensors, DC Motors) — Autonomous robot that detects and follows a line path; programmed entirely in Arduino C.
- **3D House Design** (AutoCAD) — Designed a detailed 3D model of a residential house using AutoCAD.
- **Q-Travel International Website** (HTML, CSS, JavaScript) — Responsive travel agency website deployed on Netlify.
- **Contact Book** (C++, OOP, File Handling) — Console app with full CRUD operations using file-based persistence.
- **Phase Modulation System** (Proteus) — Designed and simulated phase modulation by varying carrier phase with the message signal.
- **Automatic Street Light System** (LDR) — Light-sensitive circuit that automates streetlights to reduce energy.
- **Temperature Controlled Fan** (Proteus) — Autonomous smart cooling system implemented using a fully analog design.

TECHNICAL SKILLS

Languages: C++, Python, HTML, CSS, JavaScript **Hardware:** Arduino, ESP32, ESP8266, Circuit Analysis, Analog & Digital Electronics, IoT

Tools: Multisim, Proteus, MATLAB, AutoCAD, Git/GitHub, VMware **Concepts:** OOP, DSA, Signal Processing, Operating Systems

ACHIEVEMENTS & LEADERSHIP

- **1st Place —Project Exhibition:** Won first place for IoT-Based Smart Home Gardening System at FAST NUCES CFD.
- **1st Place — Speed Wiring Competition:** Won university-level circuit assembly competition based on speed, accuracy, and technical skill.
- **IEEE Student Chapter Member:** Contributed to technical workshops and professional development activities at FAST NUCES CFD.
- **Google Developer Group — 30-Day Challenge (2024):** Completed a structured month-long coding challenge.